

DN2 Gating Strategy

DN2 Gating Strategy for Dengue EF B Cell Identification

Research Question

What is the optimal gating strategy to isolate DN (IgD⁻CD27⁻) B cells and DN2-phenotype (CD21⁻CD11c⁺) B cells from dengue patient PBMCs using the curator's fixed 11-color panel?

Sources Used

- [Ansari2025 - Peripheral T Helper Subset Drives B Cell Response in Dengue](#) — the only dengue study with DN2-phenotype (CD21⁻CD11c⁺) data; primary gating template
- [Woodruff2020 - EF B Cell Responses in COVID-19](#) — 24-marker spectral reference standard for DN1/DN2/DN3
- [Wei2007 - DN Memory B Cells in SLE](#) — original DN (IgD⁻CD27⁻) definition and IgD vs CD27 quadrant gate
- [Jenks2018 - DN2 B Cells and EF Pathway in SLE](#) — formal DN2 definition (CXCR5⁻CD21⁻CD11c⁺CD19^{hi})
- [Sanz2025 - Human Atypical B Cells Overview](#) — IgD audit criterion and DN nomenclature recommendations
- [Singh2026 - DENV-Specific Memory B Cell Subsets](#) — dengue panel lacking CD11c/CXCR5 (negative example of incomplete DN resolution)

Panel

Fluorochrome	Marker	Role
RB705	CD19	B cell lineage
PE-Cy7	CD66b	Dump (granulocytes)
PE	CD11c	DN2 subgating
FITC	CD21	DN2 subgating
BV421	CD38	PB exclusion + transitional exclusion
eFluor506	L/D	Viability
BV711	CD3 + CD14	Dump (T cells + monocytes)
BV785	IgD	DN gating
APC	CD27	PB exclusion + DN gating
AF700	CD24	Transitional exclusion
APC-Fire750	CD45	Leukocyte gate

Sanz2025 IgD audit: PASS. IgD (BV785) is included, distinguishing true IgD⁻CD27⁻ (DN) cells from activated memory B cells that may lose CD27 but retain IgD.

Synthesis

Gating Hierarchy

The strategy follows Ansari et al. 2025 (dengue) as the primary template, with Woodruff et al. 2020 (COVID-19) DN subgating as the reference standard.

STEP 0a – SINGLETs

FSC-A vs FSC-H → Exclude doublets

STEP 0b – MORPHOLOGICAL GATE

FSC vs SSC → Generous gate including lymphoblast region

⚠ Do NOT draw a tight lymphocyte gate – activated B cell blasts (higher FSC/SSC) will be systematically excluded.

Alternative: skip this gate and rely on lineage markers only.

STEP 1 – LIVE LEUKOCYTES

APC-Fire750 (CD45⁺) → eFluor506 (L/D⁻)

STEP 2 – B CELL LINEAGE

BV711 (CD3⁻ CD14⁻) → PE-Cy7 (CD66b⁻) → RB705 (CD19⁺)

⚠ CD3 and CD14 antibodies must be titrated independently before pooling in BV711 channel.

STEP 3 – EXCLUDE PLASMABLASTS / ASCs

APC (CD27) vs BV421 (CD38)

→ Polygon gate around CD27^{hi}CD38^{hi} cluster (upper-right)

→ Exclude this population

⚠ Use a polygon gate hugging the PB cluster, NOT a quadrant line.
A quadrant risks losing CD27⁺CD38^{mid} activated memory B cells.

STEP 4 – EXCLUDE TRANSITIONAL B CELLS

AF700 (CD24) vs BV421 (CD38)

→ Polygon gate around CD24^{hi}CD38^{hi} cluster

→ Exclude this population

⚠ Set this gate on the POST-Step-3 population specifically.
CD38 dynamic range is compressed after PB removal.
Validate with CD10 in a separate tube if available.

STEP 5 – DN GATE (IgD vs CD27 quadrant)

BV785 (IgD) vs APC (CD27)

Naïve	Unswitched Memory
IgD ⁺ CD27 ⁻	IgD ⁺ CD27 ⁺
★ DN ★	Switched Memory

IgD ⁻ CD27 ⁻	IgD ⁻ CD27 ⁺
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→ Gate on the IgD⁻CD27⁻ (lower-right) quadrant = DN B cells

STEP 6 – DN SUBGATING (CD21 vs CD11c)

FITC (CD21) vs PE (CD11c) – within DN gate only

DN1-like CD21 ⁺ CD11c ⁻	CD21 ⁺ CD11c ⁺
DN3-like CD21 ⁻ CD11c ⁻	★ DN2-phenotype ★ CD21 ⁻ CD11c ⁺

→ DN2-phenotype = CD21⁻CD11c⁺

→ DN1-like = CD21⁺CD11c⁻

→ DN3-like = CD21⁻CD11c⁻

⚠ CRITICAL: CD11c-PE FMO control is MANDATORY on every acquisition.
CD11c is dim on B cells. PE-Cy7 (CD66b) tandem dye degradation creates spillover into PE that can produce false CD11c positivity.

⚠ CD21 FITC FMO also required – FITC is a relatively dim fluorochrome and FITC→PE spread must be compensated.

Backgating Verification

After gating, verify all subsets by backgating onto parent populations:

- Backgate DN2-phenotype cells onto FSC/SSC to confirm they fall within the morphological gate
- Backgate onto CD19 vs SSC to confirm clean B cell identity
- Backgate onto IgD vs CD27 to confirm they remain in the IgD⁻CD27⁻ quadrant

Comparison to Published Strategies

Feature	This panel	Ansari2025 (dengue)	Woodruff2020 (COVID-19)	Wei200 (SLE)
Colors	11	15+	24 (spectral)	8
CD19 ⁺ lineage	Yes	Yes	Yes	Yes
Dump channel	CD3/CD14/CD66b/L/D	CD14/CD16/CD3/Dead	Comprehensive	CD3
PB exclusion	CD27 ^{hi} CD38 ^{hi}	CD38 ⁻ CD27 ⁻ pre-enrichment	CD27 vs CD38	—
Transitional exclusion	CD24 ^{hi} CD38 ^{hi}	CD24 ⁻ CD38 ⁻	CD24 vs CD38	—

Feature	This panel	Ansari2025 (dengue)	Woodruff2020 (COVID-19)	Wei200 (SLE)
DN gate	IgD ⁻ CD27 ⁻	IgD ⁻ CD27 ⁻	IgD ⁻ (IgM ⁻)	IgD ⁻ CD27 ⁻
DN2 resolution	CD21 ⁻ CD11c ⁺	CD21 ⁻ CD11c ⁺	CD21 ⁻ CD11c ⁺ + CXCR5 ⁻	—
CXCR5	No	No	Yes	No
T-bet	No	No	Yes (intracellular)	No
IgM/IgG	No	Likely yes	Yes	Yes
IgD included	Yes	Yes	Yes	Yes
Sanz2025 IgD audit	PASS	PASS	PASS	PASS

Key comparability note: DN2-phenotype frequencies (% of CD19⁺ B cells) are directly comparable between this panel and Ansari2025. The gating order differs slightly (this panel excludes PBs and transitionals as separate steps; Ansari pre-enriches with CD38⁻CD27⁻ before IgD/CD27 quadrant), but the final DN and DN2 populations are defined identically.

Terminology: "DN2-phenotype" Not "DN2"

The formal Jenks2018 DN2 definition requires: IgD⁻, CD27⁻, **CXCR5⁻**, CD21⁻, CD11c⁺, CD19^{hi}, **T-bet⁺**, **FCRL5⁺**. This panel lacks CXCR5, T-bet, and FCRL5. The population isolated at Step 6 is therefore "DN2-phenotype" — enriched for but not formally confirmed DN2.

This is the same limitation Ansari et al. 2025 had. In SLE, Jenks2018 showed >90% concordance between CD21⁻CD11c⁺ and T-bet⁺ within the DN gate, and CD21⁻ is a strong surrogate for CXCR5⁻ (discordant fraction <5–10%). **Whether this concordance holds in dengue is unknown.**

All reporting should use "DN2-phenotype" throughout, following Ansari2025 precedent.

Council-Identified Risks and Mitigations

MAJOR: PE-Cy7 → PE Spectral Spread (Methodology Critic)

CD11c on PE is the gate-defining marker for DN2-phenotype. CD66b on PE-Cy7 can bleed into PE via tandem dye degradation. Since CD11c is dim on B cells, even modest spillover produces false positivity.

Mitigation: CD11c-PE FMO control on every acquisition. Empirically assess PE-Cy7→PE spread. Consider running a compensation validation with CD66b-PE-Cy7 single-stain at high concentration.

MAJOR: CD27 Shedding Contamination (Claims Validator)

Dengue is a high-TNF/IL-6 environment. ADAM17-mediated CD27 cleavage on activated switched memory B cells could place them in the IgD⁻CD27⁻ gate. If also CD11c⁺ upon activation, they contaminate the DN2-phenotype gate.

Mitigation: Not solvable within this panel. Must acknowledge as a limitation in all reporting. Measuring soluble CD27 in matched serum samples could quantify the extent of shedding. If sCD27 is elevated, DN frequencies should be interpreted with caution.

MAJOR: Post-PB-Exclusion CD38 Compression (Methodology Critic)

After removing CD38^{hi} plasmablasts (Step 3), the remaining CD38 dynamic range is compressed, making the CD24^{hi}CD38^{hi} transitional gate harder to resolve.

Mitigation: Always set the Step 4 gate on the post-Step-3 population. If a separate validation tube with CD10 is available, use CD10⁺ to validate the transitional gate boundary. In adult dengue PBMCs, transitional B cells are typically rare, so modest imprecision has limited quantitative impact.

MAJOR: FSC/SSC Blast Exclusion (Methodology Critic)

Tight lymphocyte morphological gates exclude activated B cell blasts (larger, more granular).

Mitigation: Draw the FSC/SSC gate generously to include the lymphoblast region. Alternatively, omit the morphological gate and rely on CD45/lineage markers for cleanup. Either approach is acceptable; the generous gate is preferred for consistency with published protocols.

What This Panel Resolves

- Total DN (IgD⁻CD27⁻) frequency and kinetics in dengue
- DN2-phenotype (CD21⁻CD11c⁺) frequency — directly comparable to Ansari2025
- DN1-like (CD21⁺CD11c⁻) and DN3-like (CD21⁻CD11c⁻) frequencies
- All four IgD/CD27 B cell quadrants (naive, unswitched memory, switched memory, DN)
- Plasmablast (CD27^{hi}CD38^{hi}) frequency
- Transitional B cell (CD24^{hi}CD38^{hi}) frequency
- Cross-disease DN2-phenotype benchmarking vs. SLE (Jenks2018) and COVID-19 (Woodruff2020)

What This Panel Cannot Resolve

Gap	Missing marker	Impact
Formal DN2 confirmation	CXCR5, T-bet, FCRL5	Must use "DN2-phenotype" terminology

Gap	Missing marker	Impact
DN1 vs DN3 distinction	CXCR5	Both are CD21 ⁺ or CD21 ⁻ but CXCR5 ⁺ vs CXCR5 ⁻
Isotype within DN2	IgM, IgG, IgA	Cannot determine class-switch status
Proliferative status	Ki-67	Cannot assess active cycling
CD27 shedding vs true DN	sCD27 or secondary marker	Cannot distinguish loss from absence
DENV antigen specificity	Tetramer / antigen probes	Cannot determine DENV-binding fraction
Long-lived PC phenotype	CD138	Cannot assess ASC maturation state

Panel Strengths (Council Assessment)

1. **Direct Ansari2025 comparability** — the only dengue DN2 reference. Generates the second-ever dengue DN2-phenotype dataset (STRONG).
2. **11-color efficiency** — achieves DN subgating with cleanup on conventional instruments. Accessible to dengue-endemic-setting labs without spectral capability (STRONG).
3. **Sanz2025 IgD audit compliant** — credible DN reporting; multiple dengue studies fail this criterion (STRONG).
4. **Comprehensive dump** — CD3, CD14, CD66b, L/D removes T cells, monocytes, granulocytes, and dead cells. CD66b inclusion is an improvement over Ansari2025 (MODERATE).
5. **CD24 transitional exclusion** — absent from most dengue B cell panels (MODERATE).

Open Questions

- Does the CD21⁻/CXCR5⁻ concordance established in SLE (Jenks2018, <5–10% discordance) hold in acute dengue? This determines whether "DN2-phenotype" approximates true DN2.
- How much CD27 shedding occurs in acute dengue? Elevated sCD27 would inflate apparent DN frequencies. No dengue study in the wiki quantifies this.
- Can a second-tube validation panel (adding CXCR5, T-bet intracellular, IgM, IgG) be run on a subset of samples to calibrate the primary panel's DN2 purity?
- Is the DN2-phenotype:DN1-like ratio informative of disease severity in dengue, as it is in SLE (Jenks2018) and COVID-19 (Woodruff2020)?
- Does the DN2-phenotype expansion track with days post-fever-onset, paralleling or preceding the plasmablast wave?

Related Pages

[Double-Negative B Cell](#), [DN2 B Cell](#), [DN3 B Cell](#), [Plasmablast](#), [CD21](#), [CD11c](#), [CD27](#), [IgD](#), [CD38](#), [CD24](#), [CXCR5](#), [T-bet](#), [FCRL5](#), [Extrafollicular Response](#), [Conventional Flow Cytometry](#), [Spectral Flow Cytometry](#), [FACS Sorting](#), [Research Plan - DN B Cell Expansion in Dengue](#)